Transpiration Stream

Exam code: 0610 & 0970

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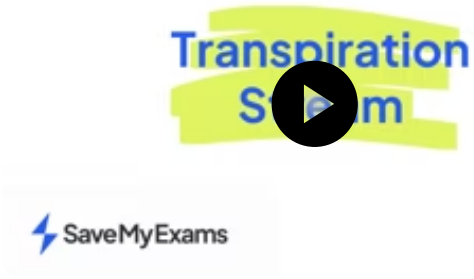
Water Vapour Loss: Extended

Extended Tier Only

- **Evaporation** takes place from the surfaces of spongy mesophyll cells
- The **many interconnecting air spaces** between these cells and the stomata create a **large surface area**
- This means evaporation can happen rapidly **when stomata are open**

Help me understand



Did this video help you? Yes No

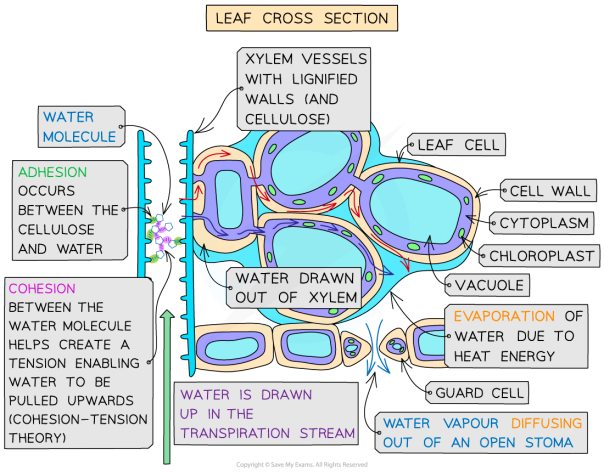
Transpiration Stream: Extended

Extended Tier Only

- Water molecules are attracted to each other by **cohesion** - creating a continuous column of water up the plant
- Water moves through the xylem vessels in a continuous **transpiration stream** from roots to leaves via the stem
- Transpiration produces a **tension** or 'pull' on the water in the xylem vessels by the leaves

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

- If the rate of transpiration from the leaves **increases**, water molecules are pulled up the xylem vessels **quicker**



The generation of the transpiration stream

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